

Introduction

1. model provides a missing link between debates on financialization and PK models that are mostly focused on the real economy.
2. the notion of speculation and liquidity preference has always been there, but whether and to what extent it has an impact is debate. while for mainstreamers money as a whole is only a veil, Keynes emphasizes that the demand for money – e.g. in the form of hoarded savings – can have real impacts. however, in treatise of money he denies that the supply of money for speculative purposes has such impact. our approach shows what happens if there is such a link and clarifies the most obvious assumptions that come with imposing one in a general framework of endogenous money.
3. our argument on turnover implicitly creates a multiplier that governs the mismatch in scale between speculative transactions and GDP (also already discussed in the Treatise!)

A simple static model

The core model

Three versions:

1. In the first one just speculation is added and it is shown how this inflates balance sheets.
2. In the second one we impose the link via c to show this has real impact. we could compare this with a scenario of rising inequality.
3. we do the same as in c with a different notion of causality, where increasing total bank balance sheets constrain lending to that the reaction of the banking sector is modeled in greater detail. should give a result similar to (b) that is better explained.

Endogenizing α - towards an endogenous asset turnover rate

Endogenizing c - credit constraints through speculation

Dual interest rate policies

Policies

→ for this we need some 2/4 standard panel to explain results.

Steps towards a dynamic model

The goal is to establish a dynamic model, that can capture hysteresis effects as well as the nonlinear dynamics of the static versions

The core model without the asset market stays unchanged

$$\begin{aligned}
Y_t &= ND_t + cD_t \\
ND_t &= bY_t \\
D_t &= d_0 - d_1 r_t \\
i_t &= i_0 + i_1 P_t \\
r_t &= (1 + m)i_t \\
dL_t &= cD_t + SD_t \\
dM_t &= dL_t \\
dR_t &= kdM_t \\
P_t &= (1 + n)aW_t \\
W_t &= W_0 - hU_t \\
N_t &= aY_t \\
U_t &= 1 - \frac{N_t}{N^f}
\end{aligned}$$

In the asset market the asset supply changes from the static model in that we partially endogenize AQ by assuming that AQ grows by the factor g_α through new assets being created. Of these new assets β are sold to investors, while $(1 - \beta)$ are held back in reserves

$$\begin{aligned}
SD_t &= s_0 - s_1 r_t \\
AD_t &= \gamma_0 + \left(\frac{1}{1 - \gamma} \right) \frac{SD_t}{AP_t} \\
AQ_t &= (1 + g_\alpha)AQ_{t-1} \\
AS_t &= \alpha AQ_t \\
AP_t &= p_1 \frac{AD_t}{AS_t}
\end{aligned}$$

or alternatively if we allow firms to keep back assets at a factor $1 - \beta$ we could adjust this to

$$\begin{aligned}
SD_t &= s_0 - s_1 r_t \\
AD_t &= \gamma_0 + \left(\frac{1}{1 - \gamma} \right) \frac{SD_t}{AP_t} \\
AQ_t &= (1 + g_\alpha)AQ_{t-1} \\
AC_t &= AC_{t-1} + \beta(AQ_t - AQ_{t-1}) - \alpha_0 AC_t \\
AF_t &= AF_{t-1} + (1 - \beta)(AQ_t - AQ_{t-1}) - \alpha_1 AF_{t-1} \\
AS_t &= \alpha_0 AC_t + \alpha_1 AF_{t-1} + \beta(AQ_t - AQ_{t-1}) \\
AP_t &= p_1 \frac{AD_t}{AS_t}
\end{aligned}$$

Following the same procedure as the static model we now try to endogenize α . We assume that the decision in which quantity to sell assets is taken on previous period prices. So the equations for α is given as

$$\alpha_t = s(AP_{t-1})$$

where s could be either linear, or a more complex function that incorporates panic selling (e.g

$$s(AP) = \frac{\omega_p}{1 + e^{k_p(AP - P_p)}} + \frac{\omega_h}{1 + e^{-k_h(AP - P_h)}}$$

Endogenizing c

In the same vain as α we can endogenize c without problem by taking the formula developed in the static model but taking the base year prices

$$c_t = c_0 - c_1 AD_t$$

Super duper extra: A energy based hysteresis model

A totally different direction could be to incorporates a energy based hysteresis model to determine the asset price

$$SD_t = s_0 - s_1 r_t$$

$$AQ_t = (1 + g_\alpha) AQ_{t-1}$$

$$AS_t = \alpha AQ_t$$

$$AP_t = \arg \min_m U(m) - \langle m, SD_t \rangle + \chi \|m - AP_{t-1}\|$$